DIVE DEEP + INTELLIGENT URBAN TRANSPORTATION

A.P.EX

Presenting UrbanDive — environmental data, made actionable.

Built on Oracle Autonomous Database 26ai

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The Problem.

The data exists. But it is scattered, buried in tables, and full of gaps. Decision-makers cannot easily see water, air, or conservation status for the places they care about.

Coverage gaps

Monitoring stations are sparse. Most locations have no direct readings.

Data silos

Urban and environmental data live in separate systems.

Inaccessible

Reading it takes SQL or a dashboard. Most stakeholders can't.

Our Solution.

An interactive map of the SF Bay. See the data by location, query it in plain English, make better decisions.

01

See the data

Color-coded points across the bay. Water, air, conservation status.

02

Ask the data

Plain English queries. No SQL, no dashboards.

03

Guide decisions

Real data behind every recommendation.

Target Audience

Three groups shape the bay. We built this for all of them.

Government

Ground policy in real environmental data.

Businesses & developers

Scout sites with environmental risk built in.

Conservationists

Back advocacy with data, not just intuition.

Starting With The Data.

Two datasets. One bay. Smart Data captures urban activity; Otter Data captures the environment. They overlap on the things that matter most — water quality, air quality, wildlife, vessels.

SMART DATA

Urban activity

- Air Quality
- Tourism Monthly
- Water Quality
- Vessel Traffic

OTTER DATA

Environmental readings

- Water quality
- Microplastic Sampling
- Monitoring Stations
- Tourism Harbor Visitors

Starting With The Data.

Water Quality Table

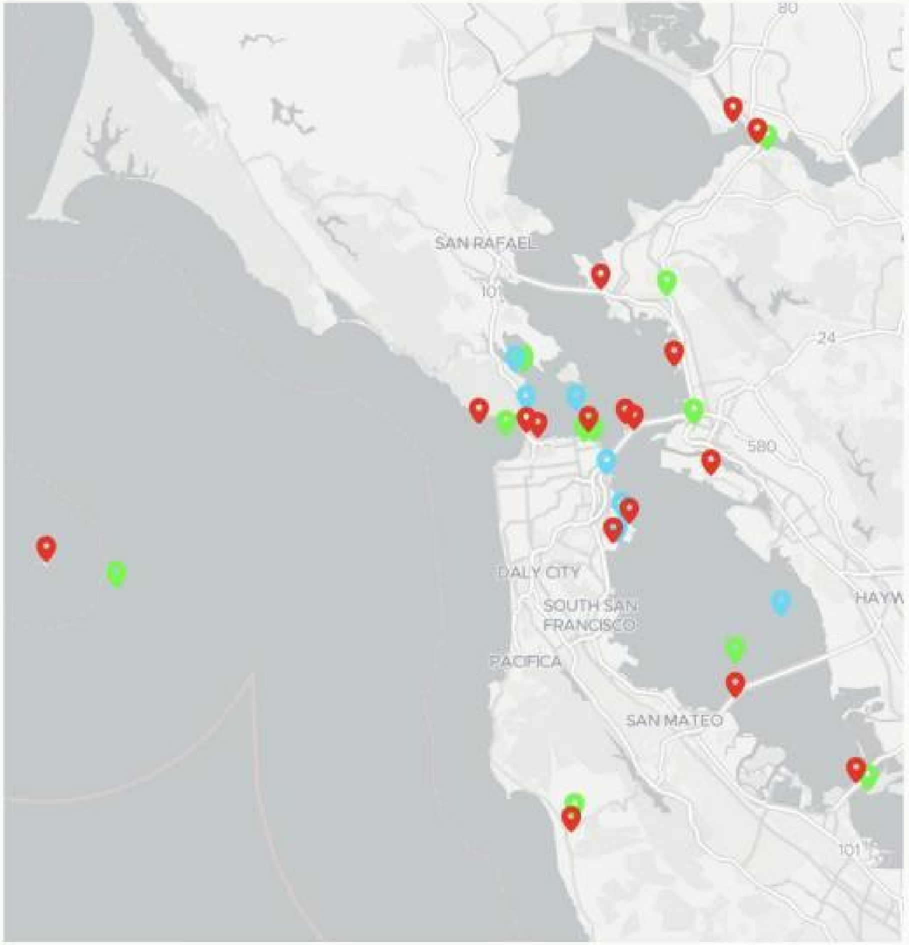
STATION_ID	STATION_NAME	LATITUDE	LONGITUDE	LOCATION_GEOM	ZONE	READING_ID	READING_TIMESTA	WATER_TEMP_CELS	WATER_QUALITY_I
STN-012	Dumbarton Bridge	37.506	-122.118	MDSYS.SDO_GEO...	Far South Bay	WQ-0082595	2025-07-08 18:00:...	20.44	0.693
STN-019	Alameda Estuary	37.771	-122.277	MDSYS.SDO_GEO...	East Bay	WQ-0114515	(null)	11.94	0.6071
STN-012	Dumbarton Bridge	37.506	-122.118	MDSYS.SDO_GEO...	Far South Bay	WQ-0082257	(null)	19.22	0.6446
STN-008	Richmond Bridge	37.931	-122.397	MDSYS.SDO_GEO...	San Pablo Bay	WQ-0053257	(null)	18.89	0.7343
STN-018	Hunters Point	37.7297	-122.366	MDSYS.SDO_GEO...	South Bay	WQ-0110884	(null)	19.71	0.7151
STN-005	Pier 39 Marina	37.8087	-122.4098	MDSYS.SDO_GEO...	Waterfront	WQ-0033975	(null)	20.27	0.7438
STN-019	Alameda Estuary	37.771	-122.277	MDSYS.SDO_GEO...	East Bay	WQ-0119394	(null)	18.8	0.7326
STN-013	Mare Island Strait	38.074	-122.253	MDSYS.SDO_GEO...	San Pablo Bay	WQ-0088596	(null)	12.92	0.5971
STN-005	Pier 39 Marina	37.8087	-122.4098	MDSYS.SDO_GEO...	Waterfront	WQ-0032389	(null)	19.62	0.7244

Microplastics Table

	STATION_ID	STATION_NAME	LATITUDE	LONGITUDE	SAMPLE_DATE	POLLUTION_INDEX
1	STN-016	Sausalito Waterfro...	37.859	-122.485	02/28/2021, 12:00:...	0.2481
2	STN-004	Treasure Island So...	37.815	-122.37	03/21/2021, 12:00:...	0.4578
3	STN-018	Hunters Point	37.7297	-122.366	03/21/2021, 12:00:...	0.3549
4	STN-006	Crissy Field Beach	37.804	-122.465	04/18/2021, 12:00:...	0.2342
5	STN-005	Pier 39 Marina	37.8087	-122.4098	05/16/2021, 12:00:...	0.3627
6	STN-019	Alameda Estuary	37.771	-122.277	08/01/2021, 12:00:...	0.2122
7	STN-010	Candlestick Point	37.713	-122.383	10/31/2021, 12:00:...	0.3076
8	STN-016	Sausalito Waterfro...	37.859	-122.485	02/27/2022, 12:00:...	0.171
9	STN-013	Mare Island Strait	38.074	-122.253	07/10/2022, 12:00:...	0.3562

The Map.

APEX reads each record's latitude/longitude and drops a color-coded point. Each color equals a data type. Click a point to see every stat the underlying table holds for that location.



LEGEND

- Water quality
- Pollution
- Conservation status

The ML Model.

We built custom ML tables from our data and trained Oracle AutoML on them. The model learns the relationships between water quality, air quality, and conservation status across the bay.

TRAINED TODAY

Model foundation

- Custom ML tables built from Smart + Otter Data
- AutoML trained on water temp, distance to Golden State, Oakland port, and San Jose
- Learns how urban activity relates to environmental outcomes

BUILDING NEXT

Spatial recommendations

- Click anywhere on the map for an insight
- Model recommendations for unmonitored locations
- Score infrastructure impact before it's built

The ML Model.

Algorithm	Model Name	R2	Mean Squared Error	Median Absolute Error
Neural Network	NN_8AB6FE84FA	0.9528	0.0002	0.0086

Features

Refresh

Filter

Name	Importance	Type
WATER_TEMP_CELSIUS		NUMBER
DIST_TO_GOLDEN_GATE		NUMBER
DIST_TO_OAKLAND_PORT		NUMBER
DIST_TO_SAN_JOSE		NUMBER
TARGET_WQI		NUMBER

The ML Model.

```
--should be in the 70s
SELECT PREDICTION(NN_8AB6FE84FA USING
    18.5 as WATER_TEMP_CELSIUS,
    75 as DIST_TO_GOLDEN_GATE,
    60  as DIST_TO_OAKLAND_PORT,
    5.0 as DIST_TO_SAN_JOSE
) AS predicted_wqi
FROM DUAL;
```

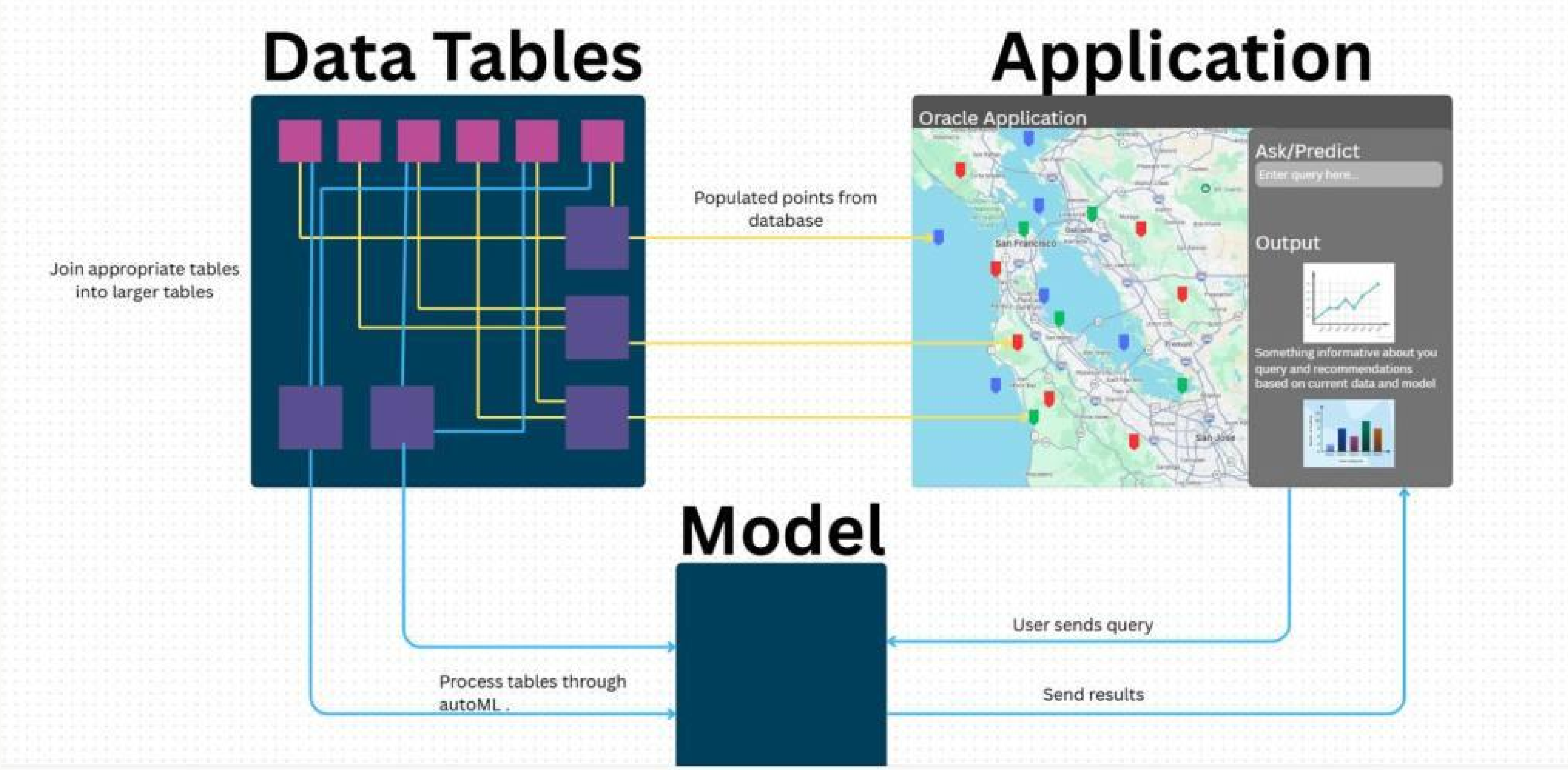
	PREDICTED_WQI
1	0.7441602396006...

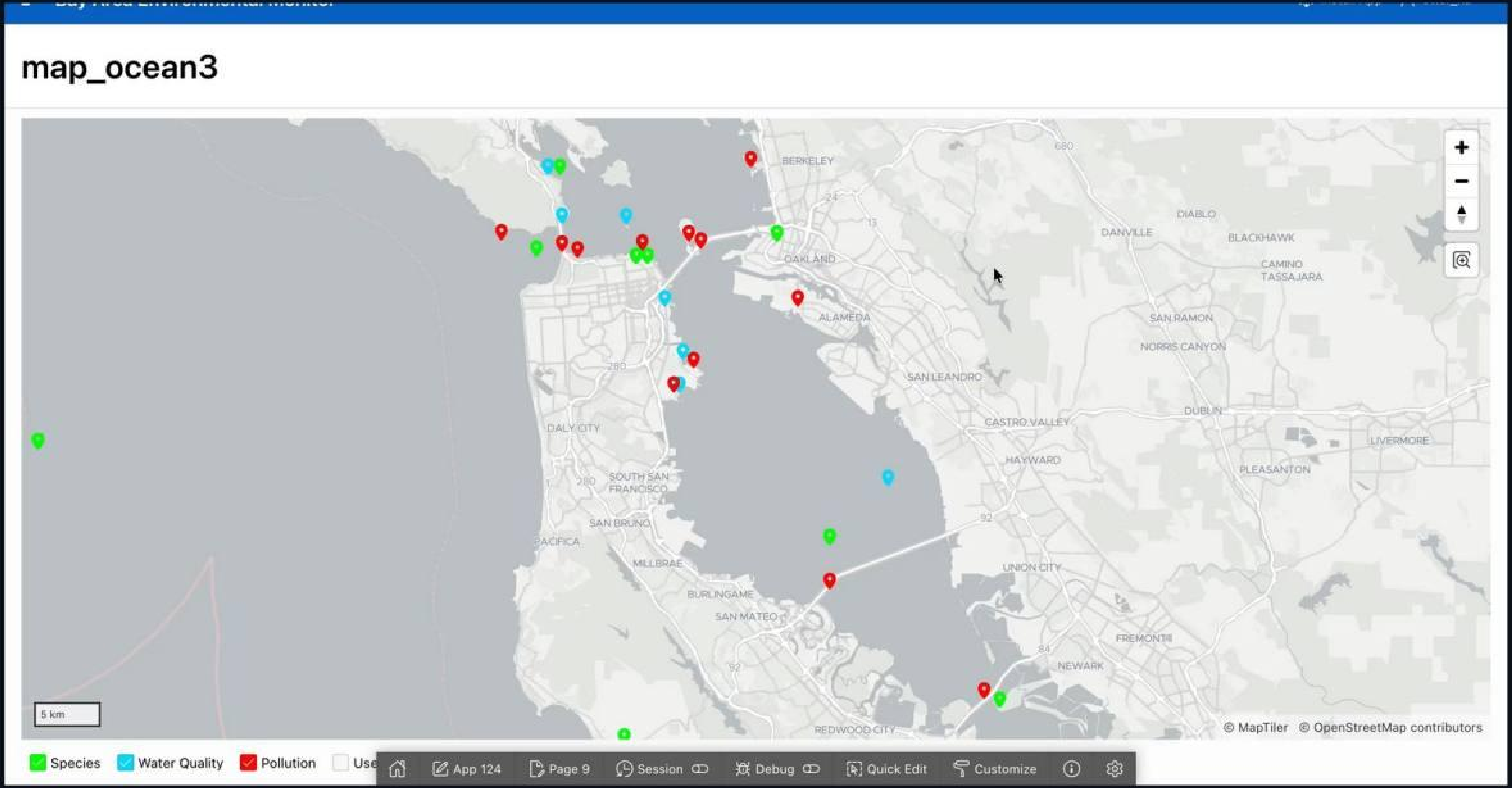
```
--should be in the 90s
SELECT PREDICTION(NN_8AB6FE84FA USING
    13.5 as WATER_TEMP_CELSIUS,
    1.5 as DIST_TO_GOLDEN_GATE,
    12  as DIST_TO_OAKLAND_PORT,
    85.0 as DIST_TO_SAN_JOSE
) AS predicted_wqi
FROM DUAL;
```

	PREDICTED_WQI
1	0.9186179812010...

Architecture.

The database is organized into parent domains; each parent owns its own tables. Location parents feed the APEX map directly. Activity parents feed the ML model. APEX queries the model live and shows the result.




map_ocean3
BERKELEY
OAKLAND
ALAMEDA
SAN LEANDRO
DANVILLE
DIABLO
BLACKHAWK
CAMINO
TASSAJARA
SAN RAMON
NORRIS CANYON
DUBLIN
CASTRO VALLEY
HAYWARD
LIVERMORE
PLEASANTON
DALY CITY
SOUTH SAN FRANCISCO
SAN BRUNO
PACIFICA
MILLBRAE
BURLINGAME
SAN MATEO
UNION CITY
FREMONT
NEWARK
REDWOOD CITY
5 km
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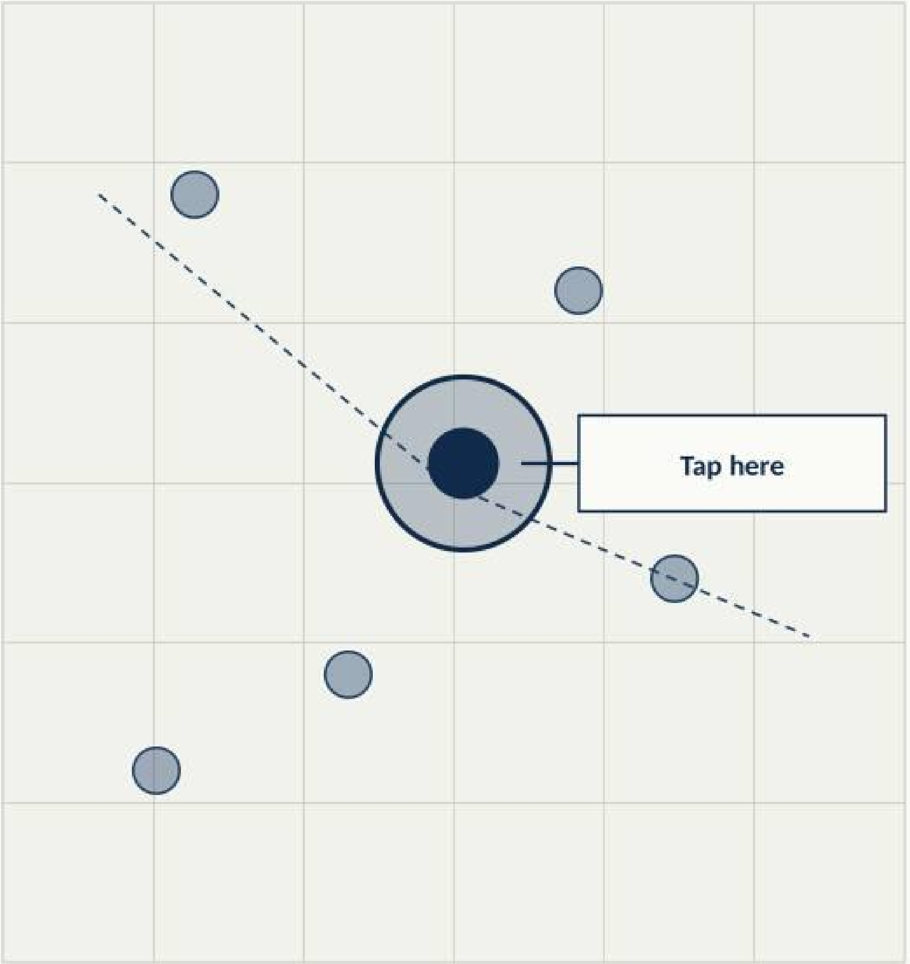
Future Plans

Next: Turn the whole map into a live query. Tap any point in the bay anywhere, not just existing stations and the model returns an environmental recommendation for that exact spot.



WHAT IT UNLOCKS

- Coverage anywhere in the bay every pixel becomes queryable
- Instant environmental context for any address
- Machine Learning Improvement/Implementation

Future Plans

Next: place a proposed building, dock, or road on the map and get an environmental impact score before construction starts. Turn "what could happen" into "here's what will happen."

WHAT IT UNLOCKS

- Test infrastructure ideas before a single permit
- Compare alternatives side-by-side on real data
- Flag high-risk proposals before they're approved

IMPACT SIMULATION

Scenario: new dock at Pier 39

	BEFORE	AFTER
	Today	**With dock**
Water	82	71
Air	78	75
Conservation	85	68

TEAM A.P.EX

Thanks.

Questions?